

3A - 4A - 7A - Kepler

Midterm Master Test Review

41 Review Questions

TEACHER EDITION

ASTRONOMY: OUR JOURNEY THROUGH THE COSMOS

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TEACHER EDITION

Test 2 Master Test Review

41 review questions in subject order, matching the student edition question for question. Each item gives a model answer, where students find it, the Canvas bank code(s), and the question-bank JSON IDs for TeacherCompose. A dash means no code exists.

Scientific Measurement

1. What are the three parts of a scientific measurement?

ANSWER Best guess, tolerance, and unit; for example, in 12.34 ± 0.02 cm the best guess is 12.34, the tolerance ± 0.02 , and the unit cm. The least count and the ILE are not parts of the statement.

Find answers in: 3AE1 · **Canvas:** T1P11 · **JSON:** 3AE1-P1-Q2

2. What is the confidence interval for 12.34 ± 0.02 cm? In percent, what probability is associated with the confidence interval of a single scientific measurement?

ANSWER 12.32 cm to 12.36 cm, endpoints included. The probability is 100%.

Find answers in: 3AE1 · **Canvas:** T1P19 · **JSON:** 3AE1-P1-Q3, 3AE1-P2-Q1

3. What is the difference between the least count and the tolerance of a measuring device? Do they always have the same value? Give an example.

ANSWER The least count is the smallest increment the device shows; the tolerance is the uncertainty attached to a reading. They are not always the same: a micrometer has a least count of 0.01 mm but a tolerance of 0.005 mm. A digital caliper has 0.01 mm for both.

Find answers in: 3AE1 · **Canvas:** T1P12C · **JSON:** 3AE1-P3-Q5, 3AE1-P4-Q3

4. How is a milliliter related to a cubic centimeter?

ANSWER They are equal: $1 \text{ mL} = 1 \text{ cm}^3$.

Find answers in: 3AE1 · **Canvas:** — · **JSON:** 3AE1-P5-Q3

5. How many significant figures are in each of these measurements, written in standard notation: 0.081704 cm and 0.0069 cm?

ANSWER 0.081704 cm has five; 0.0069 cm has two. Leading zeros are only placeholders; the zero between the 7 and the 4 is significant.

Find answers in: 3AE2 · **Canvas:** T1P12, T1P15 · **JSON:** SIGFIG-1.1, SIGFIG-1.5

6. The following is a scientific measurement written in scientific notation: 4.0560×10^3 cm. How many significant figures does this measurement have?

ANSWER Five. Trailing zeros written in scientific notation are always significant, because otherwise you wouldn't write them.

Find answers in: 3AE2 · **Canvas:** T1P13 · **JSON:** SIGFIG-2.2, SIGFIG-2.3

7. Multiply the following two measured quantities together, and state your answer to the correct number of significant figures, rounding up as necessary: 12.846×21.37 . What is your answer?

ANSWER 274.5. The calculator gives 274.51902, but 21.37 has only four significant figures, so the product keeps four.

Find answers in: 3AE2 · **Canvas:** T1P16 · **JSON:** SIGFIG-6.4

8. What is accuracy, and how is it different from precision?

ANSWER Accuracy is how close you are to the true value of a measured quantity. Precision is how finely you can measure, shown by the number of significant figures you can legitimately write.

Find answers in: 3AE2 · **Canvas:** T1P14 · **JSON:** SIGFIG-3.1, SIGFIG-3.2

9. In a scientific measurement, is being precise necessary, sufficient, or both, in order to be accurate? Explain.

ANSWER Necessary but not sufficient. You cannot be accurate without being precise, but a precise device can still be inaccurate, for example if it is miscalibrated (high precision, low accuracy).

Find answers in: 3AE2 · **Canvas:** T1P20 · **JSON:** SIGFIG-3.4, SIGFIG-3.5

10. A nontrivial systematic error is an inherent limitation in what two aspects of an experiment? Give an example.

ANSWER The equipment or the procedure. Example: starting and stopping a timer by hand in response to a voice command. (Trivial errors, such as zero error and parallax, can and should be corrected.)

Find answers in: 3AE2 · **Canvas:** T1P39 · **JSON:** SIGFIG-4.1, SIGFIG-4.3

Astronomical Distances

11. List astronomy's three distance units from smallest to largest. In miles, how big is an astronomical unit (au)?

ANSWER The astronomical unit, the light-year, and the parsec. One au is about 93 million miles.

Find answers in: 3AE3 · **Canvas:** T1P41 · **JSON:** 3AE3-P1Q2, 3AE3-P2Q1

12. When is Earth closest to the Sun? If not distance, what causes the seasons?

ANSWER At perihelion, during Northern Hemisphere winter. The seasons are caused by the tilt of Earth's axis, which changes the angle at which sunlight strikes each region.

Find answers in: 3AE3 · **Canvas:** — · **JSON:** 3AE3-P2Q2, 3AE3-P2Q3

13. If light takes about 500 seconds to travel from the Sun to Earth, how many light-seconds is one au?

ANSWER About 500 light-seconds.

Find answers in: 3AE3 · **Canvas:** — · **JSON:** 3AE3-P3Q2

- 14.** Who discovered that the period of a Cepheid variable star (bright to dim to bright) could be used to find its distance? How are its period and luminosity related?

ANSWER Henrietta Leavitt, in 1912, as one of the “Harvard computers.” The longer the period, the more luminous the star.

Find answers in: 3AE3 · **Canvas:** T1P42 · **JSON:** 3AE3-P6Q1, 3AE3-P6Q3

- 15.** What is the Chandrasekhar Limit, and why are Type Ia supernovae useful for measuring distance?

ANSWER About 1.4 solar masses. A Type Ia supernova's peak brightness is well known, especially after correcting for how quickly the light fades, so it works as a standard candle.

Find answers in: 3AE3 · **Canvas:** — · **JSON:** 3AE3-P7Q1, 3AE3-P7Q4

Gravity and Kepler's Laws

- 16.** What is a freely falling object?

ANSWER An object that moves under the influence of gravity only.

Find answers in: 4AE1 · **Canvas:** T2A4 · **JSON:** 4AE1-P1Q2

- 17.** How are weight and mass different, and how are they related?

ANSWER Mass is the amount of matter in an object; weight is the gravitational force on it. They are not identical: $W = mg$.

Find answers in: 4AE1 · **Canvas:** T2A5 · **JSON:** 4AE1-P2Q2

- 18.** Something weighs 150 pounds on Earth. How much does it weigh on the Moon, and does its mass change?

ANSWER About 25 pounds, since the Moon's surface gravity is about one-sixth of Earth's. Its mass does not change.

Find answers in: 4AE1 · **Canvas:** T1P2 · **JSON:** —

- 19.** According to Newton's Law of Universal Gravitation, the gravitational force between two masses is proportional to what?

ANSWER The product of the two masses (and inversely proportional to the square of the distance between their centers).

Find answers in: 4AE1 · **Canvas:** T1P26 · **JSON:** 4AE1-P2Q3

- 20.** If the distance between the centers of two masses doubles, what happens to the gravitational force between them?

ANSWER It becomes one-fourth as large, because the force depends on $1/\text{distance}^2$ and $2^2 = 4$.

Find answers in: 4AE1 · **Canvas:** — · **JSON:** 4AE1-P2Q4

- 21.** State Newton's Third Law of Motion.

ANSWER For every action, there is an equal and opposite reaction.

Find answers in: 4AE1 · **Canvas:** T1P25 · **JSON:** 4AE1-P3Q5

22. At which lunar phases do spring tides and neap tides occur?

ANSWER Spring tides: New Moon and Full Moon. Neap tides: First Quarter and Third (Last) Quarter.

Find answers in: 4AE1 · **Canvas:** T1P27 · **JSON:** 4AE1-P3Q7

23. Is Earth's measured acceleration due to gravity larger at the Equator or at the North Pole? State the approximate values.

ANSWER Larger at the North Pole, about 9.832 m/s^2 , than at the Equator, about 9.780 m/s^2 .

Find answers in: 4AE1 · **Canvas:** T1P30 · **JSON:** 4AE1-P4Q3, 4AE1-P4Q4

24. What did Kepler find about the shape of planetary orbits?

ANSWER All planetary orbits around the Sun are ellipses, with the Sun at one focus.

Find answers in: Kepler exercise · **Canvas:** T2A6 · **JSON:** 4AE1-P5Q4

25. What does Kepler's Second Law say, and what does it mean for a planet's speed?

ANSWER Planets sweep out equal areas in equal amounts of time. A planet speeds up as it approaches perihelion and moves slowest at aphelion.

Find answers in: Kepler exercise · **Canvas:** T2A7, T2A7A · **JSON:** KEPLER01-3, KEPLER01-5

26. List the planets in order from the Sun outward.

ANSWER Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune.

Find answers in: 4AE1 · **Canvas:** T2A2 · **JSON:** —

27. What is Pluto?

ANSWER A dwarf planet.

Find answers in: 4AE1 · **Canvas:** T2A1 · **JSON:** —

Our Cosmic Neighborhood

28. Why do astronomers use the 24-hour clock and UTC? How far behind UTC is a city at 90° W?

ANSWER The 24-hour clock removes AM/PM ambiguity, and UTC, kept at the Prime Meridian in Greenwich, avoids time zones and Daylight Saving Time. Earth turns 15° per hour, so 90° W is 6 hours behind. Chicago is near 90° W: on standard time, midnight in London is 6 pm the previous day in Chicago.

Find answers in: 7AE1 · **Canvas:** T3P52A · **JSON:** 7AE1-P1Q1, 7AE1-P1Q3, 7AE1-P1Q4

29. What does equinox mean?

ANSWER "Equal night." It is the moment the Sun crosses the Equator, when day and night are about equal everywhere on Earth.

Find answers in: 7AE1 · **Canvas:** — · **JSON:** 7AE1-P2Q3

- 30.** Where does the Sun rise and set on an equinox, and how is the setting Sun's path related to an observer's latitude? What is Chicago's latitude?

ANSWER Due east and due west. The angle between the setting Sun's slanted path and the vertical, where it meets the horizon, equals the observer's latitude. Chicago is at about 42° N.

Find answers in: 7AE1 · **Canvas:** T3A53 · **JSON:** 7AE1-P3Q1, 7AE1-P3Q4

- 31.** Which three stars form the Winter Triangle?

ANSWER Betelgeuse, Sirius, and Procyon.

Find answers in: 7AE1 · **Canvas:** T3A38 · **JSON:** 7AE1-P5Q1

- 32.** How do you find Orion, and which star is his left shoulder as he faces away from you?

ANSWER Find the belt of three bright, evenly spaced stars. Orange-red Betelgeuse is the left shoulder.

Find answers in: 7AE1 · **Canvas:** T3A38 · **JSON:** 7AE1-P6Q1

- 33.** How many moons does Mars have, and what are they called? What gives Mars its red color?

ANSWER Two, Phobos and Deimos. Its red color comes from iron oxide (rust).

Find answers in: 7AE2 · **Canvas:** T3A39 · **JSON:** 7AE2-P1Q2, 7AE2-P1Q3

- 34.** What is the tallest volcano in the Solar System, and where is it?

ANSWER Olympus Mons, on Mars.

Find answers in: 7AE2 · **Canvas:** T3A56 · **JSON:** —

- 35.** Why is Venus seen only as a morning or evening star? Give its ancient names.

ANSWER It is an inferior planet, orbiting inside Earth's orbit, so it never appears far from the Sun. Evening star: Hesperus (Greek) and Vesper (Roman); morning star: Phosphorus (Greek) and Lucifer (Roman).

Find answers in: 7AE2 · **Canvas:** T3A41 · **JSON:** 7AE2-P2Q1, 7AE2-P2Q2

- 36.** Why does the surface of Venus stay near 867 °F, both day and night?

ANSWER A runaway greenhouse effect in its dense carbon dioxide atmosphere, about 92 times Earth's surface pressure, traps the heat.

Find answers in: 7AE2 · **Canvas:** T3A36 · **JSON:** 7AE2-P2Q3

- 37.** How did the Moon form, according to the giant-impact model? How far away is it now?

ANSWER A Mars-sized body, Theia, struck the early Earth a glancing blow, and the debris formed the Moon. It is now about 239,000 miles away on average.

Find answers in: 7AE2 · **Canvas:** T2A3 · **JSON:** 7AE2-P3Q1

- 38.** What does it mean that the Moon is tidally locked, and is its far side always dark?

ANSWER Its rotation period equals its orbital period, so it always shows the same side to Earth. The far side is not always dark; it gets sunlight just as often as the near side.

Find answers in: 7AE2 · **Canvas:** — · **JSON:** 7AE2-P4Q1

- 39.** Distinguish a sidereal month from a lunation.

ANSWER A sidereal month, about 27.3 days, is measured against the stars. A lunation (synodic month), about 29.5 days, runs from New Moon to New Moon.

Find answers in: 7AE2 · **Canvas:** T3A47 · **JSON:** 7AE2-P4Q2

- 40.** List the Moon's phases in order, starting from New Moon. Which phase comes between New Moon and First Quarter?

ANSWER New, waxing crescent, First Quarter, waxing gibbous, Full, waning gibbous, Third (Last) Quarter, waning crescent. Waxing crescent comes between New Moon and First Quarter.

Find answers in: 7AE2 · **Canvas:** T3A42 · **JSON:** 7AE2-P4Q3

- 41.** What kind of galaxy is the Milky Way? Name the five largest members of the Local Group.

ANSWER A barred spiral galaxy. The five largest Local Group members are the Andromeda Galaxy (M31), the Milky Way, the Triangulum Galaxy (M33), the Large Magellanic Cloud, and the Small Magellanic Cloud.

Find answers in: 7AE2 · **Canvas:** T3A40 · **JSON:** 7AE2-P5Q1, 7AE2-P5Q3
